AI Use Appendix

The Italian Stagnation: A Multi-Lens Visual Investigation

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Tool used. Anthropic's Claude was the only AI tool used in this project. No image-generation tools, no other large language models, and no IDE autocomplete plug-ins beyond standard syntax highlighting were used.

Purposes. We used Claude only in the limited assistant role defined by the project brief. Specifically:

- *Debugging.* When a notebook raised an unexpected error, we sometimes pasted the traceback and asked for a likely cause.
- *Finding documentation.* Lookups for Plotly polar layout, Eurostat indicator codes, NUTS-2 conventions, LaTeX commands.
- *Generating boilerplate code.* The initial matplotlib styling block, the LaTeX preamble of this report, and the first scaffolding of the D3.js calculator.
- *Improving wording.* Asking for shorter or clearer rewrites of paragraphs we had already drafted.
- *Suggesting chart ideas.* Brainstorming when we were torn between two visual forms (most notably for T3, where we compared sunburst, bubble scatter, heatmap and Coxcomb).

Every analytical decision in the project (the central research question, the choice of datasets, the visual forms we kept, the editorial structure of the article, the white-hat / black-hat levers, the findings we feature) was made by the group. Claude was never used to invent data, to validate findings, or to write the project's conclusions.

Two representative prompts.

1. "We have employment-share and labour-productivity broken down by firm-size class for four European countries, and we want a single chart that shows both dimensions at once. Which chart families fit, and what are the trade-offs of each?" The reply listed sunburst, bubble scatter, heatmap, polar/Coxcomb. We prototyped the first three, decided we did not like them, and built the Coxcomb ourselves.
2. "This Plotly polar chart in our notebook renders with the title slightly overlapping the first ring. What's the standard way to add top margin without breaking the layout?" The reply pointed to the `margin=dict(t=...)` layout key. We applied it, verified the output, kept it.

What we accepted and what we rejected. We accepted suggestions that we could verify or test directly: chart-type proposals we then prototyped on real data, code snippets we read line-by-line before running, wording rewrites we re-edited to match the article's register. We rejected, on principle and consistently: (a) any claim involving specific numerical values we had not verified against the dataset, (b) any methodological caveat that introduced a cause we could not actually see in the data, and (c) prose written in AI-typical register (em-dashes used as separators, empty abstractions, generic newspaper-style metaphors), which we rewrote in our own voice.

One example of AI output we corrected. Early in the drafting of the T4 caption, Claude proposed a methodological caveat attributing part of the senior wage decline to the "2011 Fornero pension reform's compositional effect on the 50+ band:" the idea being that the reform raised retirement ages, kept lower-paid 60–64 workers in the active population, and dragged the senior median down. The claim is plausible in principle but it is *not* visible in our ISTAT RACLI data, and ISTAT's own documentation makes no such note. We treated it as unverified speculation and removed every reference to the Fornero reform from the project: from the paper, from the figure caption on the website, from the matplotlib annotation in `Task_4.ipynb`, from the design document, and from the per-task SOURCES file. The artefact now documents that the senior band lost the most in real-wage terms without speculating about the cause.

Accountability. The group remains responsible for every sentence, figure, code block, visual decision and conclusion in the submission. AI accelerated work we could have done ourselves; it did not replace any of the reasoning.